

CB-YOLO-DeepSORT : Real-Time Yield Estimation for Tomatoes in Greenhouses

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Abstract—To address the challenge of maturity monitoring and fruit quantity estimation in tomato yield prediction under greenhouse conditions, this paper proposes the CB-YOLO-DeepSORT framework for continuous tracking and dynamic yield estimation. Built upon YOLOv5s, the framework incorporates the CBAM attention mechanism to enhance tomato maturity recognition in occluded environments and integrates the CIoU loss function to improve localization accuracy, ensuring precise ripeness mapping for reliable yield estimation. Then, the DeepSORT-based multi-object tracking algorithm is applied to continuously monitor tomatoes at different maturity stages, maintaining identity consistency to enhance the stability of real-time tomato yield estimation. Validation confirms the enhanced CB-YOLO attains 99.5% mAP@0.5 on test data, surpassing YOLOv5s by 2.5 percentage points. The proposed system achieved MOTA and MOTP scores of 82.35% and 84.5% in medium-density scenarios, demonstrating robust tracking performance under moderate target congestion. Finally, a web-based monitoring system deploys autonomous robots with depth cameras to track tomato maturity in greenhouses. Real-time data is visualized through daily snapshots (pie charts and counts of ripe/unripe fruits) and weekly maturation trends (line charts and histograms), providing actionable insights to optimize harvest scheduling and resource allocation for precision agriculture.

Key words: Tomato maturity recognition, Multi-object tracking, Yield estimation, Inspection platform

I. INTRODUCTION

Tomatoes, as globally significant economic crops, have been widely cultivated in greenhouses due to the environmental control advantages they offer. To achieve automation and precision management in agricultural production, growth monitoring, maturity assessment, and yield estimation are essential components for greenhouse tomatoes. However, current greenhouse tomato inspection primarily relies on manual labor, which suffers from inefficiency, high costs, and substantial subjective errors, failing to meet the demands of large-scale cultivation management [1]. In recent years, deep learning-based object detection algorithms such as YOLO have demonstrated promising performance in agricultural visual monitoring tasks. Nevertheless, the complex greenhouse environment imposes additional constraints on the robustness of visual algorithms, while persistent challenges remain in small target recognition, dynamic

occlusion scenarios, and maturity classification accuracy.

Existing research has achieved notable progress in tomato fruit maturity recognition. Zhang et al. [2] developed a maturity detection method combining machine vision with tactile sensing technology using transfer learning strategies. Miao et al. [3] proposed an optimized object detection approach based on improved YOLOv7 to address issues of fruit occlusion and indistinct maturity feature differences. Hao et al. [4] introduced GSBF-YOLO, a lightweight detection model incorporating GSim, C3Ghost, and BiFPN structures to reduce computational costs, while employing FocalEIou Loss to enhance detection accuracy in occlusion scenarios. Li et al. [5] developed the LOX-SE-GIoU model, which integrates SE attention modules and achieves a 92.17% mAP in distinguishing tomato fruits from stems. Despite these advancements in detecting tomatoes at various maturity stages, limitations persist regarding small target detection and maturity classification stability in complex greenhouse environments.

Accurate counting of tomatoes at different maturity stages is crucial for intelligent yield estimation [6]. Zhang et al. [7] proposed a tomato counting method using ultra-deep masking and improved YOLOv8, enhancing counting accuracy through optimized convolutional structures and robust loss functions. An et al. [8] developed StraTracker, a multi-object tracking algorithm combining YOLOv8n detectors with feature association and dual-region counting modules for strawberry counting. Zhou et al. [9] achieved precise strawberry counting with an 8.7% error rate by integrating MultiMap algorithms with attention mechanisms. He et al. [10] introduced a feature cascade matching-based multi-object counting method to optimize fruit tracking and counting stability in dense occlusion scenarios.

While these studies have advanced fruit detection and counting techniques, they exhibit limitations in addressing complex greenhouse environments and lack real-time monitoring capabilities for fruit quantity and yield estimation, thus failing to meet smart agriculture's requirements for dynamic inspection and intelligent management. This paper proposes a greenhouse tomato yield estimation method based on CB-YOLO object detection and the DeepSORT multi-object tracking

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algorithm. The method optimizes the YOLOv5 algorithm by incorporating the CBAM attention mechanism to enhance recognition capability in complex backgrounds for tomatoes at different ripening stages. Additionally, the integration of the DeepSORT tracking algorithm improves the stability of tracking and counting. Based on this approach, an intelligent tomato inspection and yield monitoring platform has been developed to enable real-time detection of greenhouse tomatoes, providing the tomato inspection robot with precise management capabilities.

II. MATERIALS AND METHODS

A. Dataset Collection and Annotation

Tomato fruit images were collected from the Jingpeng Smart Greenhouse at the Agricultural Machinery Research Institute in Tongzhou District, Beijing, and a greenhouse tomato plantation in Rongcheng, Weihai, Shandong Province. To enhance the diversity of the dataset, images were captured under various lighting conditions Fig. 1(a. b. c), different shooting angles Fig. 1(d. e. f), and multiple illumination directions Fig. 1(g. h. i). Additionally, complex occlusion scenarios, including fruit occlusion, leaf occlusion, and stem occlusion, as shown in Fig1(j. k. l), were specifically targeted for data collection. 2116 high-quality images in sRGB format were collected, with resolutions of 4928×3264 pixels and 4032×3024 pixels, respectively. A representative set of greenhouse tomato images captured under complex environmental conditions is shown in Figure 1.



Figure 1. Images of Greenhouse Tomatoes in Complex Environments

To achieve high-precision object detection, the LabelImg annotation tool was used to manually label the pixel regions of tomato fruits in the RGB images [11]. Based on maturity stages, the tomatoes were classified into two categories:ripe tomatoes(ripe_tomatoes), characterized by a fully red surface with a firm texture, and unripe tomatoes(unripe_tomatoes), which appear dark green and remain hard. Each tomato was labeled with a minimum

enclosing bounding box to ensure accurate localization while minimizing background interference. Partially visible tomatoes were annotated only if at least 50% of their area was visible. This dataset, incorporating diverse environmental conditions and occlusion scenarios, provides reliable training data for tomato detection models.

B. Data Preprocessing and Image Augmentation

To improve the model's adaptability to complex environments and enhance dataset diversity, preprocessing and data augmentation were applied to the raw images. Due to the large image size and memory constraints, cropping, compression, and resizing were performed. Due to the large image size and memory constraints, cropping, compression, and resizing were applied [12]. The images were converted to JPEG format with a standardized resolution of 640×640 pixels, balancing computational efficiency and adaptability to low-quality images in real-world greenhouse scenarios.

In order to mitigate illumination variations, adaptive histogram equalization and random saturation adjustment ($\pm 5\%$) were applied to improve color consistency and enhance the contrast between fruits and the background. To accommodate variations in fruit growth postures, scaling, horizontal/vertical flipping, and $\pm 90^\circ$ rotations were performed, alleviating sample distribution imbalance. Additionally, Gaussian noise and 0.5-pixel Gaussian blur were introduced to mitigate overfitting, while multi-scale transformations were employed to improve the model's robustness to varying object sizes and distances.

After preprocessing and augmentation, the dataset was expanded to 4260 images, containing 14995 labeled instances, of which 9858 were "ripe_tomatoes" and 5137 were "unripe_tomatoes." The dataset was split into training (80%), testing (10%), and validation (10%) subsets, ensuring that the validation and test sets remained independent to maintain objective model evaluation. The dataset distribution across different maturity stages is presented in Table I.

TABLE I. DATASET OF TOMATO FRUITS AT DIFFER

Maturity	Training Set	Validation Set	Test Set	Total
Unripe_tomatoe	7810	1046	1002	9858
Ripe_tomatoes	4126	487	524	5137
Total	11936	1533	1526	14995

C. YOLOv5 Object Detection Model

YOLOv5 [13] demonstrates the best inference speed and recognition performance within the YOLO series. Particularly, the YOLOv5s model, due to its smaller weight and fast inference speed, exhibits stable performance and is capable of meeting the real-time and accurate recognition requirements for tomatoes at different maturity stages [14]. Therefore, in this study, we select YOLOv5s as the base model for our experiments.

III. TOMATO MATURITY CLASSIFICATION AND YIELD ESTIMATION

A. Enhanced Model with CBAM Attention Mechanism

To overcome the challenges posed by the high visual similarity between unripe green tomatoes and foliage backgrounds, as well as the loss of small-object features in greenhouse tomato images, this study proposes an improved model based on the YOLOv5 architecture, referred to as CB-YOLO. The CB-YOLO model integrates the Convolutional Block Attention Module (CBAM) and the Complete Intersection over Union (CIoU) loss function to enhance detection performance. The architecture of the improved YOLOv5 model is illustrated in Figure 2. In order to mitigate the loss of small-object features in complex backgrounds, the CBAM dual-channel attention mechanism is embedded after the CSP1_X and CSP2_1 modules in the backbone network. The CBAM module performs adaptive feature refinement through channel attention and spatial attention mechanisms.

The channel attention mechanism calculates channel-wise attention weights by applying global average pooling and max pooling to the input feature maps. The formulation is as follows:

$$M_C(F) = \sigma(W_1(W_0(F_{avg}^c)) + W_2(W_0(F_{max}^c))) \quad (1)$$

Where $M_C(F)$ represents the channel attention output weights, σ denotes the sigmoid activation function, F_{avg}^c is the feature map obtained after spatial global average pooling, and F_{max}^c is the feature map obtained after spatial max pooling. Additionally, W_0 and W_1 represent the weight matrices of the first and second fully connected layers, respectively.

The spatial attention mechanism computes spatial attention weights by applying pooling operations along the channel dimension of the feature maps. The formulation is as follows:

$$M_S(F) = \sigma(f^{7 \times 7}([F_{avg}^s, F_{max}^s])) \quad (2)$$

Where $M_S(F)$ represents the spatial attention output weights, and $f^{7 \times 7}$ denotes a 7×7 convolutional filter. Additionally, F_{avg}^s is the feature map obtained after channel-wise average pooling, while F_{max}^s is the feature map obtained after channel-wise max pooling.

To address the limitations of the original GIoU Loss [15], this study introduces the CIoU Loss [16] as a metric for measuring the deviation of predicted bounding boxes. CIoU Loss effectively mitigates the issue where GIoU Loss fails to distinguish relative positions when the predicted box is entirely contained within the ground truth box. The calculation formula is as follows:

$$L_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \quad (3)$$

Where:

$$v = \frac{4}{\pi^2} \left(\arctan \frac{\omega^{gt}}{h^{gt}} - \arctan \frac{\omega}{h} \right)^2 \quad (4)$$

$$a = \frac{v}{1 - IoU + v} \quad (5)$$

Where b and b^{gt} represent the center points of the predicted box and the ground truth box, respectively. The parameter ρ denotes the Euclidean distance between b and b^{gt} , while c represents the diagonal length of the minimum bounding rectangle enclosing both the predicted and ground truth boxes. The weight coefficient is denoted as α , and IoU refers to the Intersection over Union between the predicted and ground truth boxes. Additionally, ω^{gt} and h^{gt} denote the width and height of the ground truth box, while ω and h represent the width and height of the predicted box, respectively.

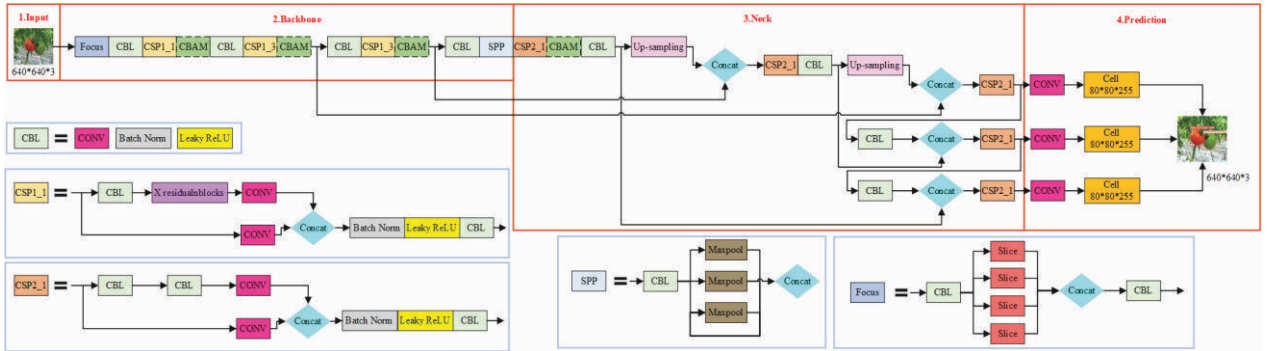


Figure 2. Improved network model with CBAM attention

B. Yield Estimation Model Based on CB-YOLO and DeepSORT

DeepSORT [17] is a multi-object tracking method based on the SORT algorithm. By introducing cascade

matching and deep learning feature extraction, it optimizes motion and appearance features, effectively reducing ID switching issues for tomatoes in long-term occlusion scenarios [18].

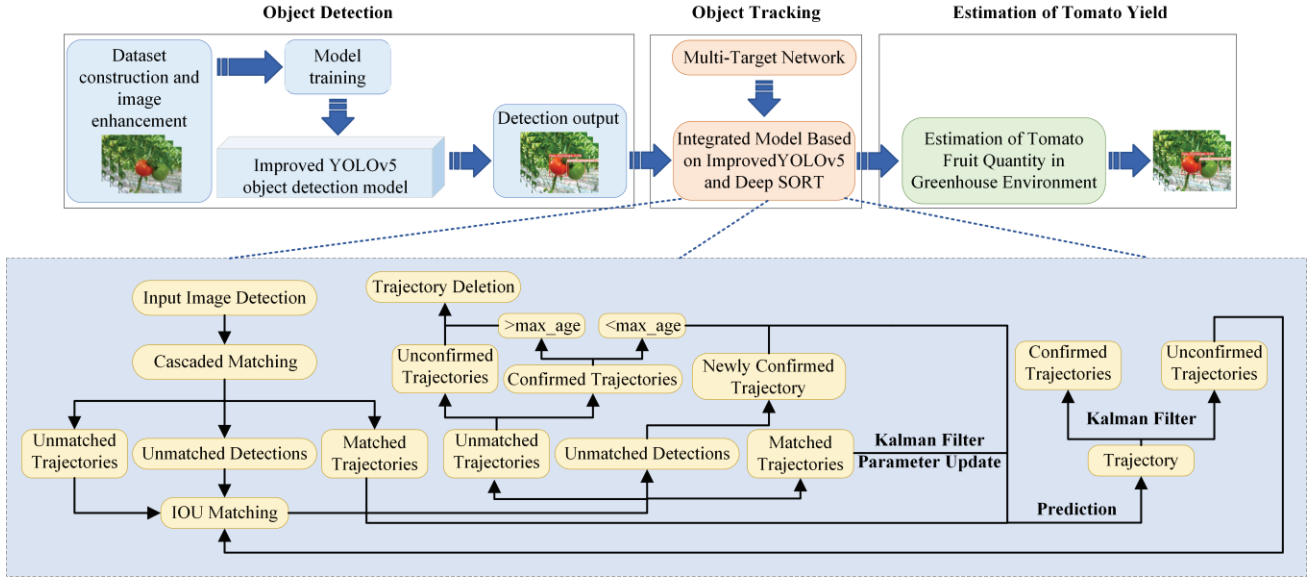


Figure 3. CB-YOLO and DeepSORT Integrated Counting Workflow

Considering the complex factors in greenhouse environments, such as wind influence, robotic arm collisions, and gravity-induced branch load changes, this paper proposes a tomato tracking and counting strategy combining CB-YOLO and DeepSORT, as shown in Figure 3. The strategy uses CB-YOLO to detect the number of tomatoes in each frame and applies DeepSORT for inter-frame tracking, ensuring consistent tomato identity and accurate counting, providing reliable data support for yield estimation.

The proposed method consists of the following key steps:

1. Online Object Detection: A high-quality tomato image dataset is constructed by collecting and augmenting tomato images. The enhanced dataset is then used to train the improved CB-YOLO detection network to obtain the optimal detection weight model.

2. Multi-Object Tracking: The improved CB-YOLO detection model is integrated with the DeepSORT tracking model to construct a multi-object tracking network. This network leverages the Kalman filter algorithm [19] and the Hungarian matching algorithm [20], significantly enhancing tomato fruit tracking performance in complex environments.

3. Tomato Yield Estimation: The improved CB-YOLO and DeepSORT models are applied to detect and track fully ripened tomatoes in video streams, enabling accurate fruit counting for yield estimation. During tracking, each detected tomato is assigned a unique ID, starting from 1, ensuring ID consistency across consecutive video frames. Thus, the maximum ID value in the video provides an accurate estimate of the total tomato yield.

C. Web-based Tomato Inspection Platform

An automated monitoring framework enables web-controlled robots to conduct daily greenhouse inspections, transmitting real-time tomato maturity analytics and yield estimates to a centralized dashboard. By combining remote control with visual feedback (ripeness percentages, harvest-ready quantities), the system reduces manual labor and

improves harvest predictability through data-driven crop management.

The system consists of three components: robot hardware, backend service module, and frontend interface. The robot hardware, based on ROS Noetic and Jetson Xavier, performs motion control and environmental perception, supporting synchronized localization and mapping for stable tomato detection, even in complex occlusion scenarios. The backend service, running on Ubuntu with the Python Flask framework, communicates with ROS via rosbridge and stores maturity and yield estimation data in an SQLite database. The frontend interface, built with HTML5 or JavaScript and roslibjs, provides an interactive control panel for robot control and displays tomato maturity and yield trends.

The integration of the improved CB-YOLO algorithm with DeepSORT enhances tomato detection and tracking, ensuring precise fruit counting. The overall platform architecture is illustrated in Figure 4.

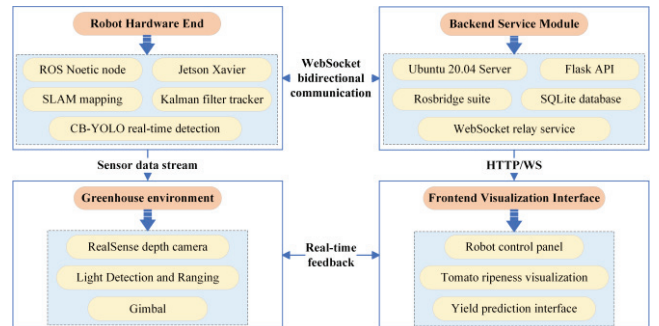


Figure 4. Tomato Inspection Platform System Architecture

IV. TEST RESULTS ANALYSIS AND INTERFACE DISPLAY

A. Construction of the Test Platform

The model training platform in this study is built on the Windows 10 operating system, with hardware

configuration consisting of an Intel I9-12900KF processor and an NVIDIA GeForce RTX 3090 24GB GPU. The software environment utilizes the CUDA 11.1 acceleration library and Python 3.7.2. The robot hardware is deployed on the NVIDIA Jetson Xavier embedded platform, running the Ubuntu 20.04 operating system and ROS Noetic framework. Before the experiments, hyperparameters were optimized. By observing the loss function curve, it was found that the model's loss gradually converged after 200 training epochs. Therefore, the training duration was set to 200 epochs for this experiment. Detailed environment configurations and experimental hyperparameters are presented in Table II.

TABLE II. MODEL PARAMETERS

Attributes	Values
Learning rate scheduling	Cosine annealing
Maximum learning rate	0.01
Num_classes	2
Number of iterations	200
Optimizer	Adam
Batch size	16

B. Ablation Study

For the improved CB-YOLO model, Precision (P), Recall (R), and mean Average Precision (mAP@0.5) are used as evaluation metrics for the recognition of tomatoes at different maturity stages. In terms of model detection performance, detection speed, and model memory usage are selected as the evaluation indicators. The specific calculation methods for these metrics can be found in the literature [21,22].

An ablation study was conducted to validate the effectiveness of the improved YOLOv5 model. The study involved adding the CBAM attention mechanism and the CIOU loss function to the original YOLOv5 model to verify the effectiveness of each improvement. The details of the ablation study are presented in Table III, where "√" indicates that the improvement was applied, and "-" indicates that the improvement was not applied.

TABLE III. ABLATION STUDY

CBAM	CIOU	Precision rate (P, %)	Recall rate (R, %)	Mean Average Precision (mAP@0.5, %)
-	-	98.4	96.4	99.3
√	-	99.3	99.5	99.5
-	√	98.8	97.0	99.3
√	√	99.8	99.5	99.5

As shown in Table III, the introduction of the CBAM attention mechanism alone improves the accuracy by 0.9 percentage points, the recall by 3.1 percentage points, and the mean Average Precision (mAP@0.5) by 0.2 percentage points, significantly outperforming the effect of introducing the CIOU loss function alone. The experiments show that the collaborative optimization of CIOU and CBAM can balance feature selection and localization accuracy, with a

performance gain that significantly exceeds that of a single module, thus validating the necessity of multi-dimensional optimization.

C. Tomato Fruit Tracking and Counting

For the DeepSORT algorithm, the primary evaluation metrics include Multi-Object Tracking Accuracy (MOTA), Multi-Object Tracking Precision (MOTP), Identity Switches (IDS).

To verify the feasibility of the proposed CB-YOLO and DeepSORT integrated algorithm for tomato tracking and counting in a greenhouse environment, experimental analysis was conducted across different scale scenarios. The tracking performance results of the tomato fruits are shown in Table IV.

TABLE IV. TRACKING AND COUNTING RESULTS ANALYSIS

Video ID	Ground Truth	Misidentification Count	Number of Unrecognized Items	IDS	MOTA/%	MOTP/%
1	8	0	0	1	87.5	86.2
2	17	1	0	2	82.35	84.5
3	33	2	1	3	81.82	84.5
4	52	3	3	4	80.77	83.9
5	89	5	6	6	80.9	83.1

The experimental results show that in low-density scenarios (where the number of tomatoes is relatively small and the fruits are widely spaced), MOTA reaches 87.5% and MOTP is 86.2%. In medium-density scenarios (where the tomatoes are more densely packed with some overlapping targets), MOTA remains around 82%, demonstrating high tracking accuracy and robustness. However, as the number of tomatoes increases, MOTA slightly decreases, possibly due to increased occlusions leading to undetected tomatoes or ID switches caused by overlapping targets. Additionally, in complex backgrounds or under significant lighting variations, the detection model may produce false positives or missed detections, affecting overall tracking performance. Nevertheless, the method maintains high accuracy across different test scenarios, validating its applicability in greenhouse environments.

D. Platform Interface Function Validation

As shown in Figure 5, the tomato inspection platform adopts an integrated interactive design, combining remote robot control, navigation, camera control, fruit detection and data analysis modules into a closed-loop management system. The system interface displays real-time data, including the total number of tomatoes, maturity distribution, estimated daily yield, and weekly maturity trends. The data is visualized through daily snapshots, including pie charts and counts of ripe and unripe fruits, as well as weekly maturity trends, displayed using line charts and histograms. The platform estimates the number of tomatoes that will ripen within 7 days by monitoring the changes in their maturity in real-time and correlating the fruit's color, thus forecasting future yield. The platform also supports precise robot motion control and multi-angle detection, enabling comprehensive monitoring and intelligent inspection operations within the greenhouse environment.

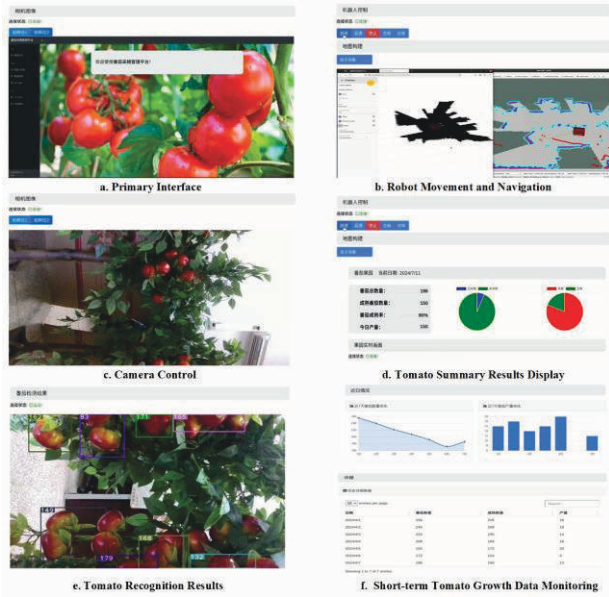


Figure 5. Screenshot of the Tomato Inspection Platform

V. CONCLUSION

This paper addresses the challenges of tomato fruit detection and yield estimation in complex greenhouse environments by proposing an integrated CB-YOLO and DeepSORT algorithm. An inspection system was developed for tomato maturity classification, tracking, and yield estimation in greenhouse settings. The key findings are as follows:

1) By embedding the CBAM attention mechanism into the YOLOv5s backbone network, the ability to distinguish between tomato fruits and the background was significantly improved. Coupled with the CIOU loss function, the issue of small object localization bias was resolved, achieving a mean average precision (mAP@0.5) of 99.5%.

2) The DeepSORT algorithm is employed, utilizing a dual matching strategy of appearance features and motion models to maintain ID consistency even after brief occlusions of the fruit. The MOTA reaches 87.5% in low-density scenarios, effectively ensuring tracking continuity.

3) The inspection platform, built on the ROS framework, successfully realized tomato maturity grading, real-time counting, and yield visualization. The practical applicability of the algorithm was validated in a real greenhouse environment, providing technical support for dynamic agricultural robot inspections.

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